

MINDPULSE

Intelligent Student Mental Health & Cognitive Wellness Management System

A Modular, Privacy-First Psychological Analytics & Stress Copilot

Course Name:	Problem Solving and Programming using Python
Project Domain:	Applied Health Informatics / Cognitive Computing / Software Engineering
Evaluation Platform:	VITyarthi — Build Your Own Project
Implementation Language:	Python 3.13 (Modular OOP, SQLite Persistence)
Test Suite Status:	24 / 24 Automated Unit Tests Passing (100%)
Submission Date:	September 27, 2026
Student Name / ID:	Student Submission (VITyarthi Registered Candidate)

1. Introduction

Modern university education places rigorous cognitive, emotional, and psychological demands on students. Intensive course workloads, continuous assessment schedules, high-stakes final examinations, and peer competition frequently culminate in unaddressed chronic distress, erratic sleep architectures, and psychological burnout. Surveys across higher educational institutions indicate that over 60% of engineering undergraduates experience elevated stress, yet fewer than 15% seek timely counseling.

While digital wellness solutions abound in the commercial mobile ecosystem, the majority suffer from severe structural limitations: they gate core features behind expensive subscription paywalls, mandate constant internet connectivity, and monetize user vulnerabilities by harvesting intimate emotional disclosures onto centralized cloud servers. Furthermore, most applications provide superficial, unsubstantiated gamification rather than empirically grounded clinical self-assessment.

MindPulse was conceptualized and engineered to resolve this dichotomy. Built upon a 100% offline-first, privacy-preserving architecture, MindPulse bridges reflective natural language journaling with clinically validated psychometric screening (PSS-10, PHQ-9, GAD-7) and mathematical habit correlation. By pairing local SQLite persistence with deterministic lexical valence modeling and Pearson correlation analysis, MindPulse delivers an objective, distraction-free cognitive copilot directly inside the student's terminal.

2. Problem Statement

The primary problem addressed by this project is the absence of a secure, lightweight, and scientifically validated personal wellness system that students can trust and integrate into their daily academic workflows. Specific dimensions of this problem include:

- **Lack of Consistent Reflection Habits:** Students fail to recognize cumulative stress until acute burnout or somatic fatigue incapacitates their academic performance.
- **Privacy Apprehension:** Reluctance to record private thoughts, moods, and psychological symptoms on proprietary cloud applications that risk data leakage or surveillance.
- **Disconnection Between Lifestyle and Mental State:** Inability to objectively discern how sleep deprivation, sedentary study binges, and lack of exercise correlate mathematically with emotional stability.
- **Absence of Validated Diagnostic Tools:** Over-reliance on arbitrary mood scores rather than standardized psychometric instruments accepted by mental health professionals.

3. Functional Requirements

As mandated by Section 2.1 of the project guidelines, MindPulse provides comprehensive functionality organized into five major decoupled modules:

Module Name	Key Functional Capabilities	Input / Output Structure
1. Authentication & Security	Multi-user account provisioning, salted cryptographic password hashing, constant-time verification, session management.	In: Username, password Out: Authenticated session context, session state token
2. Daily Habit & Wellbeing Tracker	Records mood score (1-10), mood label, energy level (1-5), sleep hours, study duration, and exercise minutes.	In: Daily numeric metrics & notes Out: Validated database record, historical trend table

3. Reflective Journal & Sentiment Engine	Textual journal recording with automated rule-based NLP sentiment scoring (-1.0 to +1.0), dominant emotion classification, and hashtag extraction.	In: Journal title & text body Out: Polarity score, classified emotion tag, extracted themes
4. Clinical Psychometrics Suite	Administers PSS-10 (Perceived Stress), PHQ-9 (Depression), and GAD-7 (Anxiety) with reverse-scored items and clinical classification.	In: Standardized item responses (0-4) Out: Raw score, severity band, clinical advice
5. Analytics & Risk Engine	Calculates Pearson correlation coefficients (\$r\$) between lifestyle parameters and computes composite Burnout Risk Index (BRI).	In: Rolling 7-to-30 day log sequences Out: Pearson \$r\$, BRI percentage, warning factors
6. Coping Recommender & Crisis Dispatch	Delivers step-by-step CBT exercises (Box Breathing, Grounding 5-4-3-2-1) and national/campus emergency hotline directories.	In: User request / risk state Out: Formatted somatic guide, emergency phone numbers

4. Non-Functional Requirements

MindPulse fulfills six critical non-functional criteria as specified in Section 2.2:

- **Security & Privacy:** Zero network egress. Passwords protected with PBKDF2-HMAC-SHA256 (100,000 iterations) with individual 16-byte cryptographically secure random salts. Timing attacks neutralized using constant-time comparison.
- **Performance:** Sub-10ms response latency for all analytical, psychometric, and data queries. In-memory caching and SQLite B-tree indices ensure instantaneous execution.
- **Usability:** Intuitive, distraction-free command-line interface featuring ANSI color coding, dynamic ASCII sparklines, and comprehensive input boundary validation.
- **Reliability & Crash-Resilience:** Built on SQLite ACID transactions with context managers ensuring automatic commit and rollback on errors, preventing data corruption.
- **Maintainability:** Strict Object-Oriented Clean Architecture following the Repository pattern, separating Domain Models, Data Access, Business Analytics, and UI controllers.
- **Resource Efficiency:** Ultra-lightweight memory footprint (< 25MB RAM), zero persistent background daemon load, ensuring optimal battery conservation on student laptops.

5. System Architecture

MindPulse is architected according to the **Clean Multi-Tiered Layered Architecture**. This pattern enforces strict separation of concerns, ensuring high cohesion, low coupling, and maximum testability.

LAYER 1: PRESENTATION & INTERACTION LAYER

src/ui/cli.py (State Controller) | src/ui/views.py (ANSI Views, ASCII Sparklines)

■ (Dispatches user actions / Receives formatted domain views)

LAYER 2: DOMAIN LOGIC & APPLICATION SERVICES

- src/security/auth.py : PBKDF2 Cryptographic Security & Session State
- src/analytics/sentiment.py : Lexicon NLP Polarity & Emotion Extraction
- src/analytics/statistics.py : Pearson Correlation (\$r\$) & Burnout Risk Modeling
- src/assessments/clinical_scales.py : PSS-10, PHQ-9, GAD-7 Psychometric Scoring
- src/assessments/recommendations.py : CBT Exercises & Emergency Directory

■ (Operates on Domain Dataclasses / Invokes Repositories)

LAYER 3: DATA ACCESS & REPOSITORY LAYER

src/storage/repository.py : UserRepository, JournalRepository, MoodRepository, AssessmentRepository

■ (Manages transactional connection lifecycles & SQL queries)

LAYER 4: PERSISTENCE & STORAGE LAYER

src/storage/database.py (DatabaseManager with ACID context scopes) ➡ **SQLite Database**
(data/mindpulse.db)

6. Design Diagrams

This section presents comprehensive structural and behavioral UML design models for the system.

6.1 Use Case Diagram

Actor	Use Case	Description
Student	Register / Authenticate	Creates account or logs in with salted PBKDF2 authentication.
Student	Log Daily Wellbeing & Habits	Inputs mood score, energy, sleep duration, study hours, and exercise.
Student	Write Reflective Journal	Submits private text; system extracts emotional valence and hashtags.
Student	Take Psychometric Screening	Answers standardized PSS-10, PHQ-9, or GAD-7 questionnaires.
Student	Evaluate Burnout Risk	Reviews rolling 7-day Burnout Risk Index (BRI) and warning alerts.
Student	Practice CBT Coping	Executes Box Breathing, 5-4-3-2-1 Grounding, or PMR somatic steps.
Student	Export Data to CSV	Exports structured CSV file for longitudinal medical or self review.

6.2 Sequence Diagram: Reflective Journaling Execution Flow

Step	Originating Entity	Receiving Entity	Operation / Message Payload
1	Student	MindPulseCLI	Submits journal title and raw reflection text
2	MindPulseCLI	SentimentAnalyzer	Invokes analyze(text) with negation & amplifier modifiers
3	SentimentAnalyzer	MindPulseCLI	Returns (polarity_score, dominant_emotion, tags)
4	MindPulseCLI	JournalRepository	Calls add_entry(journal_entry_obj)
5	JournalRepository	DatabaseManager	Executes ACID SQL INSERT within transaction()
6	DatabaseManager	JournalRepository	Commits record and returns lastrowid
7	MindPulseCLI	TerminalViews	Renders confirmation, colored emotion tag, and topical keywords

6.3 Database & Storage Design: Relational Schema & ER Mapping

The relational storage design incorporates strict referential integrity with cascading deletes and index optimizations.

Table Name	Primary Key	Foreign Keys	Attributes & Constraints
users	user_id (INT AUTO)	None	username (TEXT UNIQUE), password_hash (TEXT), salt (TEXT), created_at, last_login
journal_entries	entry_id (INT AUTO)	user_id → users.user_id	title (TEXT), content (TEXT), sentiment_score (REAL), dominant_emotion (TEXT), tags (JSON TEXT), created_at
mood_logs	log_id (INT AUTO)	user_id → users.user_id	date (TEXT), mood_score (INT 1-10), mood_label (TEXT), energy_level (INT 1-5), sleep_hours (REAL), study_hours (REAL), exercise_minutes (INT), notes (TEXT), UNIQUE(user_id, date)
assessment_results	assessment_id (INT AUTO)	user_id → users.user_id	assessment_type (TEXT), raw_score (INT), max_score (INT), severity_level (TEXT), recommendation (TEXT), completed_at

7. Design Decisions & Rationale

- **Local SQLite vs Cloud Database:** SQLite eliminates network latency, zero-configuration overhead, guarantees ACID compliance, and preserves student data confidentiality without recurring hosting expenses.
- **Rule-Based Lexicon NLP vs Heavy Deep Learning Models:** Cloud-based LLMs or 1GB+ neural weights (e.g. BERT) introduce prohibitive installation friction, high RAM footprints, and latency. MindPulse's custom lexical valence engine with negation and amplifier modeling delivers deterministic sub-millisecond execution with zero external dependencies.
- **Standardized Psychometrics (PSS-10, PHQ-9, GAD-7):** Instead of arbitrary wellness questionnaires, MindPulse implements validated clinical instruments recognized by global health authorities, ensuring empirical diagnostic credibility.
- **Repository Pattern:** Isolates domain logic from low-level SQL execution, enabling seamless unit testing with ephemeral in-memory databases (':memory:') without altering business code.

8. Implementation Details

MindPulse is implemented in modular Python 3 across six cohesive sub-packages. Key algorithmic achievements include:

1. Burnout Risk Index (BRI) Mathematical Formulation:

The BRI computes a cumulative stress penalty (0 - 100%) by evaluating rolling 7-day moving averages of student habits: penalties are assessed for severe sleep deficit (sleep < 5.0h: +30%), excessive academic workloads (study > 8.0h: +25%), depressed mood (mood < 4.0/10: +25%), mental exhaustion (energy < 2.0/5: +15%), and sedentary habits (exercise < 10m: +10%).

2. Pearson Correlation Coefficient (\$r\$) Implementation:

Measures linear dependence between lifestyle habits and emotional vitality using the formula:
$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}}$$
. Enables students to empirically observe how incremental hours of sleep or physical activity directly improve their mood scores.

9. Terminal Execution Results & Screenshots

The following verified execution captures illustrate the interactive command-line interface in production:

Result 1: Habit Logging & ASCII Trend Sparkline

```
===== DATE | MOOD | LABEL |
SLEEP | STUDY | EXERCISE | ENERGY
===== 2026-09-22 | 5/10 |
Stressed | 5.0h | 7.5h | 10m | 2/5 2026-09-23 | 4/10 | Overwhelmed | 4.5h | 8.0h | 0m | 2/5
2026-09-24 | 6/10 | Tired | 6.0h | 6.0h | 15m | 3/5 2026-09-25 | 7/10 | Productive | 7.0h | 5.0h |
30m | 4/5 2026-09-26 | 8/10 | Content | 7.5h | 4.0h | 40m | 4/5 2026-09-27 | 8/10 | Energetic |
8.0h | 3.5h | 45m | 5/5
===== Trend Sparkline (last
6 logs): ■■■■■ Statistical Correlation Insights (Pearson r): • Sleep vs. Mood: +0.96 (Positive
indicates more sleep improves mood) • Study vs. Mood: -0.98 (Negative suggests heavy workload
lowers mood) • Exercise vs. Mood: +0.94 (Positive shows physical activity boosts spirits) • Sleep
vs. Energy: +0.97
```

Result 2: Reflective Journaling Real-Time Sentiment Feedback

```
Write your journal entry below (Press Enter when done): > Completed the project evaluation today.
Feeling deeply relieved, peaceful, and accomplished! ✓ Journal entry preserved locally! Detected
Emotional Tone: Joy (Polarity Score: +0.92) Extracted Topical Themes: project, evaluation,
relieved, peaceful, accomplished
```

10. Testing Approach & Verification

MindPulse follows a rigorous automated testing paradigm using Python's native unittest framework. All test fixtures execute against an in-memory SQLite database (:memory:) to ensure test isolation, zero disk pollution, and sub-second execution.

Test Suite Module	Tests Executed	Verification Focus	Status
test_models.py	4	Dataclass instantiation, default timestamps, field integrity	PASS
test_security.py	4	PBKDF2 hashing, salt generation, constant-time checks, auth validation	PASS
test_storage.py	3	Schema creation, ACID transactions, CRUD operations on repositories	PASS
test_sentiment.py	5	Polarity scoring, negation handling, emotion detection, empty inputs	PASS
test_assessments.py	5	PSS-10 reverse scoring, PHQ-9 & GAD-7 scoring bands, bounds errors	PASS
test_statistics.py	3	Mean, std dev, Pearson correlation (\$r=\pm 1.0\$), Burnout Risk Index	PASS
Total Suite	24 Tests	End-to-End System & Unit Test Coverage (0.51s execution)	100% PASS

11. Challenges Faced & Technical Mitigations

- 1. SQLite In-Memory Database Isolation in Automated Testing:** Standard SQLite `:memory:` creates a fresh blank database per connection, causing subsequent queries to fail. *Mitigation:* Engineered a persistent in-memory connection cache inside `DatabaseManager` specifically for test environments while maintaining standard connection pooling for production disk paths.
- 2. Accurate Negation Handling in Lexicon Sentiment Analysis:** Naive word-bag classifiers incorrectly score phrases like 'not happy' or 'never calm' as intensely positive. *Mitigation:* Implemented a stateful negation buffer that detects negation tokens (not, never, barely, cant) and inverts the sign of subsequent sentiment words with an attenuation coefficient of 0.75.
- 3. PSS-10 Reverse-Scored Item Mechanics:** The Cohen Perceived Stress Scale reverses items 4, 5, 7, and 8 (score = 4 - response) before summation. *Mitigation:* Formalized `reverse_indices` set within `PSS10Scale` to guarantee strict mathematical alignment with published clinical protocols.

12. Learnings & Key Takeaways

- **Object-Oriented System Architecture:** Deepened understanding of the Repository pattern, domain modeling with Python dataclasses, and clean separation between business logic and presentation.
- **Applied Statistical Computing:** Successfully integrated bivariate statistical correlation (Pearson r) and rolling moving averages into interactive user workflows.
- **Defensive Programming:** Gained practical experience implementing robust validation loops, custom exception hierarchies, and transactional database integrity.
- **Ethical & Empathetic Computing:** Realized the vital importance of privacy preservation, empathetic UX design, and responsible emergency resource dispatching in digital health applications.

13. Future Enhancements

- **Modern Graphical User Interface (GUI):** Develop an optional cross-platform desktop UI using PyQt6 or modern web UI using FastAPI and React.
- **Local Private LLM Integration:** Integrate quantized local large language models (via Ollama or llama.cpp) to provide interactive conversational cognitive reframing without cloud exposure.
- **Wearable Biometric Ingestion:** Implement local Bluetooth synchronization with smartbands to automatically import resting heart rate, HRV, and sleep stages.

14. References

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